

WEAREDEVELOPERS X BAND: DARK FACTORY HACKATHON

Pocketful

Autonomous Zero-Sum Fintech Engine & Bitemporal Ledger

Synthesized 100% autonomously inside the BAND Desktop multi-agent factory without human steering. Delivering strict zero-sum conservation, SHA-256 cryptographic audit trails, and offline containerization.

 <https://pocketful-fintech.vercel.app>

 <https://github.com/fokrulanthro16-eng/pocketful-submission>

 <https://youtu.be/70vY0ZwfjpM>

BAND-er Kaj: Autonomous Factory Mechanics

50% Scoring Rubric

How BAND Desktop coordinates decoupled LLM seats without human steering.

⚡ BAND Desktop Architecture

BAND Desktop is a local-first multi-agent desktop runtime engineered by Thenvoi for dark software synthesis.

Core Primitives in Action:

- **Autonomous Room:** Isolated multi-seat workspace (room_dark_factory_pocketful_stage1).
- **Generic Mandates:** Standing behavioral contracts in mandates/containing 0% track-specific hints.
- **Three Decoupled Seats:** Systems Planner, Core Developer, and Quality Verifier operating via DeepSeek-R1.

🏭 Dark Factory Protocol

Zero human code steering. The system executes end-to-end delivery:

- **Planner Seat:** Ingests user kickoff, derives deterministic invariants, and produces SPEC.md.
- **Developer Seat:** Implements FastAPI ledger, SQLite WAL schema, and multi-stage Dockerfile.
- **Verifier Seat:** Dispatches test suites and gates code handoff through formal harness verification.

BAND-er Kaj: Live Interception & Self-Healing

AUTOMATED PROOF

Automated race-condition discovery, gate blocking, and remediation inside BAND.

Concurrency Interception

During Stage-1 synthesis, the **Verifier Seat** executed a stress harness simulating 50 concurrent transactions against Account ACC-01.

CRITICAL DEFECT INTERCEPTED:

A double-spend race condition on request #27 caused balance to leak below zero (-\$25.00).

Action: Verifier halted the release gate and returned raw failure logs to Developer.

Closed Autonomous Self-Healing Loop

Without human intervention, the **Developer Seat** parsed the defect trace and refactored the transaction core:

AUTOMATED REMEDIATION:

- Injected reentrant asyncio.Lock mutexes around mutate routes.
- Activated SQLite WAL serialized write-locking.
- Re-verified on 100 parallel workers: 100% Passed. Zero balance leakage. Gate certified!

App-er Kaj: Strict Zero-Sum Bitemporal Engine

PRODUCT INVARIANT

The production fintech application generated by the BAND factory.

Mathematical Conservation Invariant

Traditional ledgers suffer from balance drift during network lags or simultaneous authorization holds.

Pocketful enforces a mathematically bounded invariant:

$$\sum \Delta \text{Balances} + \sum \Delta \text{Holds} + \sum \Delta \text{Escrow} = 0$$

Every single credit is atomically balanced by an exact matching debit across participant accounts and platform reserves. Overdrafts return strict HTTP 400 rejections.

Idempotency & Bitemporal Auditing

- **Mandatory Idempotency Keys:** Every mutation header is hashed; repeated requests return cached results with zero duplicate deductions.
- **Bitemporal Timeline:** Distinguishes between transaction event-time (when payment occurred) and ledger assertion-time (when state was written).
- **Group Payment Splits:** Multi-party expense allocation conserved atomically.

App-er Kaj: Cryptographic Audit & AML Sentinel

COMPLIANCE READY

Real-time forensic tamper-proofing and pre-settlement fraud heuristics.

SHA-256 Cryptographic Hash Chaining

Every financial state mutation is chained to the preceding block:

```
Hash_n = SHA256(Hash_(n-1) || Tx_Payload || UTC_Timestamp)
```

Provides institutional auditors with tamper-proof mathematical certainty against retroactive database alterations or ledger rewrites.

Real-Time AML Sentinel Heuristics

Monitors transaction flow prior to balance commitment:

- **Velocity Anomaly Detection:** Flags rapid micro-transfers exceeding safe thresholds.
- **Structuring Heuristics:** Intercepts repeated transactions structured just below compliance reporting limits (\$10,000).
- **Automated Risk Isolation:** Quarantines suspicious payloads into escrow.

Verification Matrix & Deliverables

193/193 PASSED

Demonstrating complete rubric compliance and production deliverables.

 **193 / 193 Tests Passed (Contiguous Regression-Free)**

38/38

STAGE 1: LEDGER CORE

47/47

STAGE 2: HOLD ESCROW

52/52

STAGE 3: SPLIT PAYMENTS

56/56

STAGE 4: AML & AUDIT



Zero-Network Offline Container

Multi-stage Docker container (Python 3.11-slim) validated on port 8080 with 0 outbound network egress.



Submission Artifacts

Live App: pocketful-fintech.vercel.app

Video Demo: <https://youtu.be/70vY0ZwfjpM>